

The State of OpenRail

Quarterly Metrics Report - 2026Q3

Scope: 2026-06-01 to 2026-08-31

Issue date: 2026-09-08

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Executive Summary

The theme of Q3 2026 is broadening participation. Activity across the incubated projects remains at the healthy level of previous quarters, with OSRD continuing to account for the bulk of contributions, driven primarily by SNCF.

This quarter marks the arrival of MscViewer as a newly onboarded project, contributed by Greenstones. It brings a new contributing organization into the ecosystem and extends OpenRail's scope into railway telecommunication tooling (GSM-R).

Hack4Rail, held in June, left a visible mark on the data. Work started at the hackathon flowed into the incubated projects over the following weeks, including a new optimization layer for the DAC Migration DSS PopUpSim from Deutsche Bahn and automatic-layout work on Netzgrafik-Editor from SBB. This cross-company engagement continues the trend of collaboration beyond the projects' originating organizations.

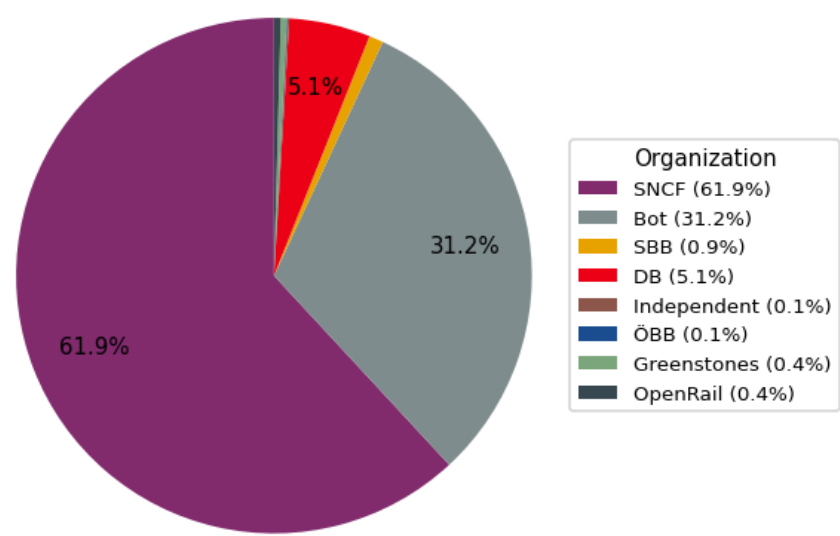
Netzgrafik-Editor sustains the elevated activity noted last quarter, reflecting the ongoing SNCF and SBB collaboration. The administrative projects remain the second-largest area of activity, supporting the association's growing operational needs.

1. Quarterly Snapshot

Summary Statistics (2026-06-01 to 2026-08-31):

- Active contributors this quarter: 67
- Human commits this quarter: 1905
- Contributing organizations this quarter: 8

Commits by Organization (This Quarter)



Code-committing organizations: DB, Greenstones, OpenRail, SBB, SNCF, ÖBB. A small number of commits were contributed by independent developers.

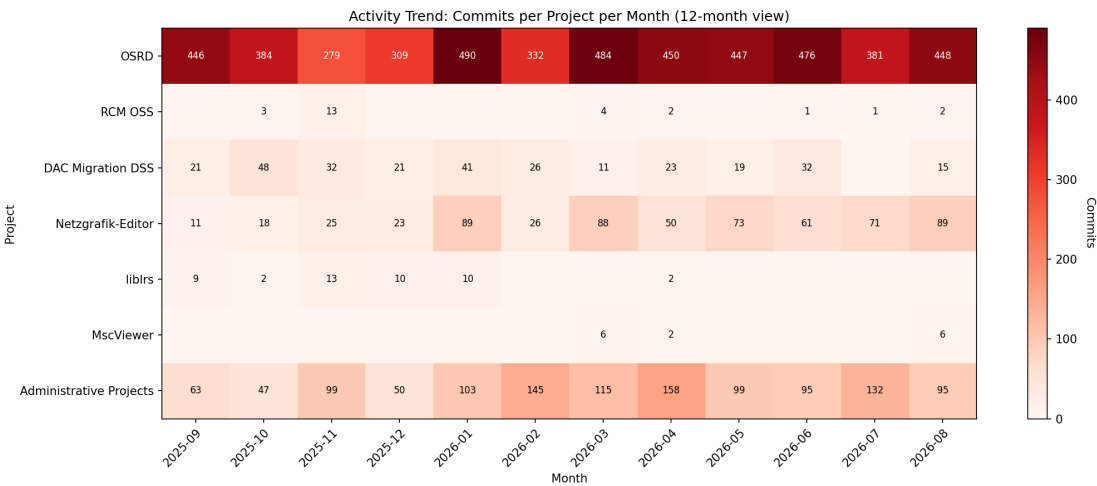
Note: Commits from automation (CI, version updates) are marked as Bot in the diagrams.

2. Progress Data (12-Month View)

This section shows activity trends over the past 12 months for context (2025-09-01 to 2026-08-31).

Overall Activity:

- Total contributors (12 months): **113**
- Total commits (12 months): **7126**
- Total organizations (12 months): **9**



Activity Trend: Commits per Project per Month

3. Project Details

Activity trends and organization contributions for each project over the past 12 months.

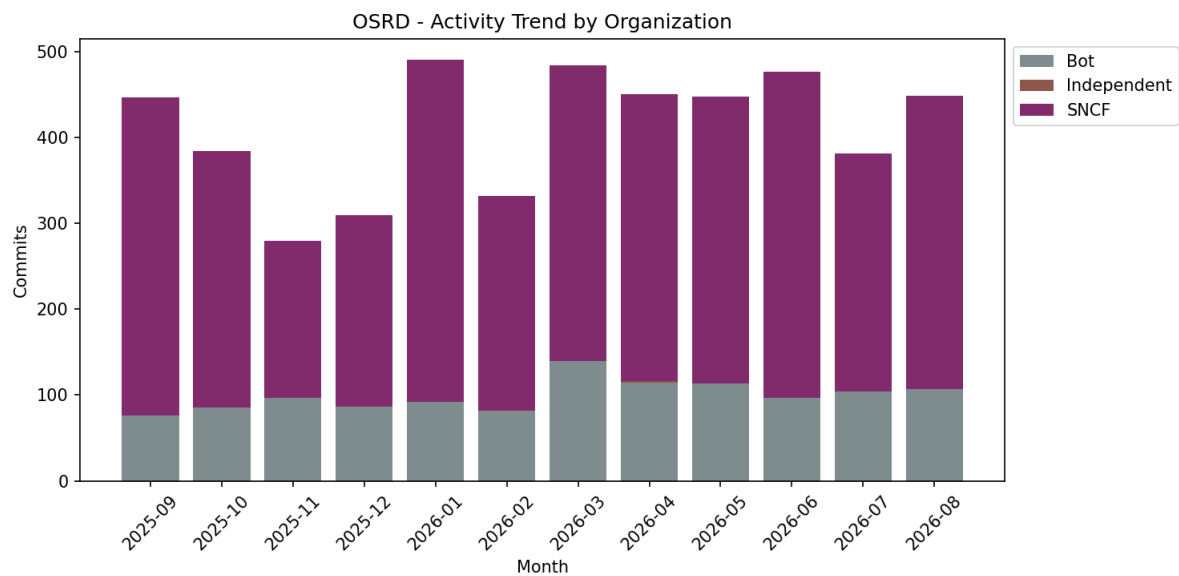
Stage 2 - Qualified

OSRD

OSRD (Open Source Railway Designer) is an open source web application for railway infrastructure design, capacity analysis, timetabling and simulation. It provides tools for designing railway networks, analyzing capacity constraints, and optimizing train schedules in a collaborative environment.

12-Month Statistics:

- Total commits: **4926**
- Total contributors: **81**
- Contributing organizations: **3**



OSRD - 12-Month Activity Trend by Organization

Repositories:

- [osrd](#)
- [osrd-design](#)
- [osrd-website](#)

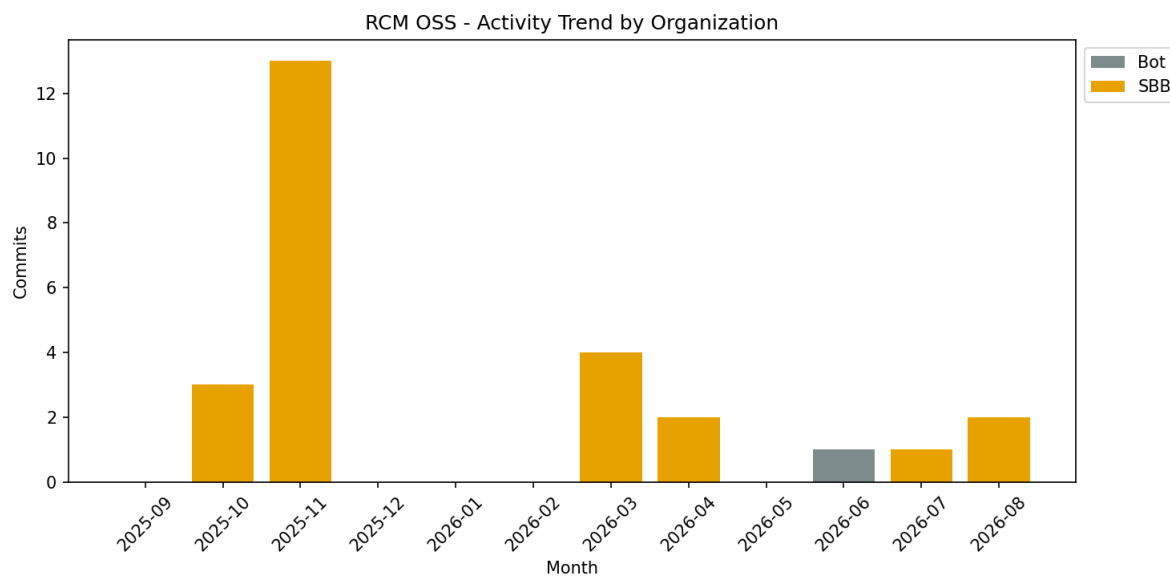
Stage 1 - Onboarded

RCM OSS

RCM-DX (Rail Condition Monitoring Data eXchange) is an open data format specification for railway diagnostic data. Based on the HDF5 format, it enables easy accessibility and visualization of railway condition monitoring data from mobile and stationary measuring systems, facilitating data exchange between railway companies and suppliers.

12-Month Statistics:

- Total commits: **26**
- Total contributors: **4**
- Contributing organizations: **2**



RCM OSS - 12-Month Activity Trend by Organization

Repositories:

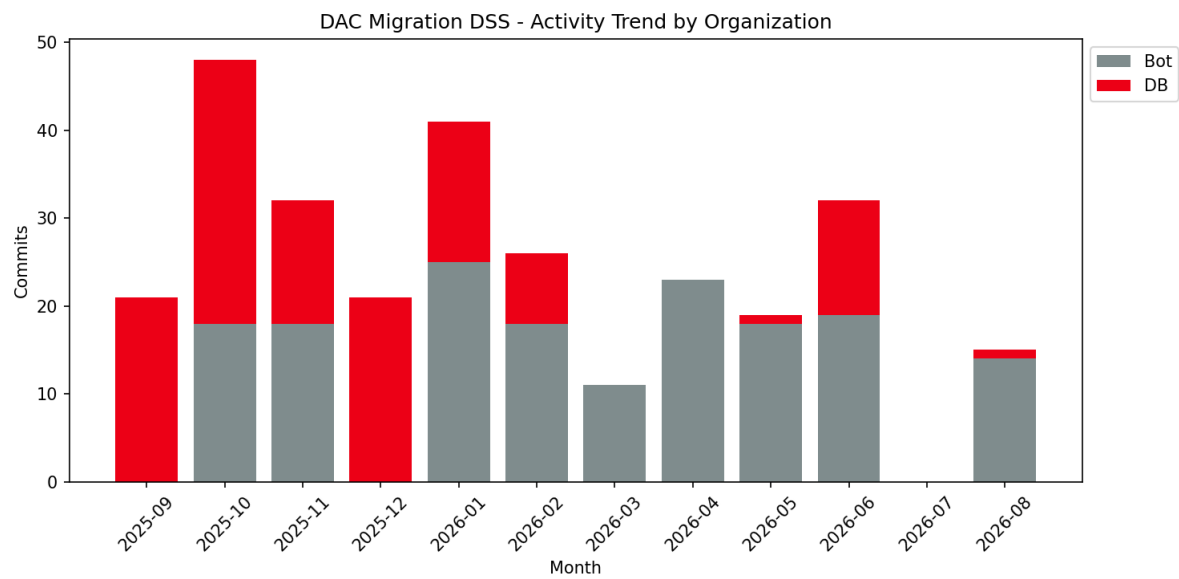
- [rcm-dx](#)
- [rcm-dx-examples](#)

DAC Migration DSS

The Digital Automatic Coupler Migration Decision Support System helps plan the retrofit of approximately 500,000 European freight wagons from screw couplers to digital automatic couplers. This strategic planning tool supports rail undertakings, wagon owners, and workshops in coordinating this unprecedented industry-wide migration.

12-Month Statistics:

- Total commits: **289**
- Total contributors: **7**
- Contributing organizations: **2**



DAC Migration DSS - 12-Month Activity Trend by Organization

Repositories:

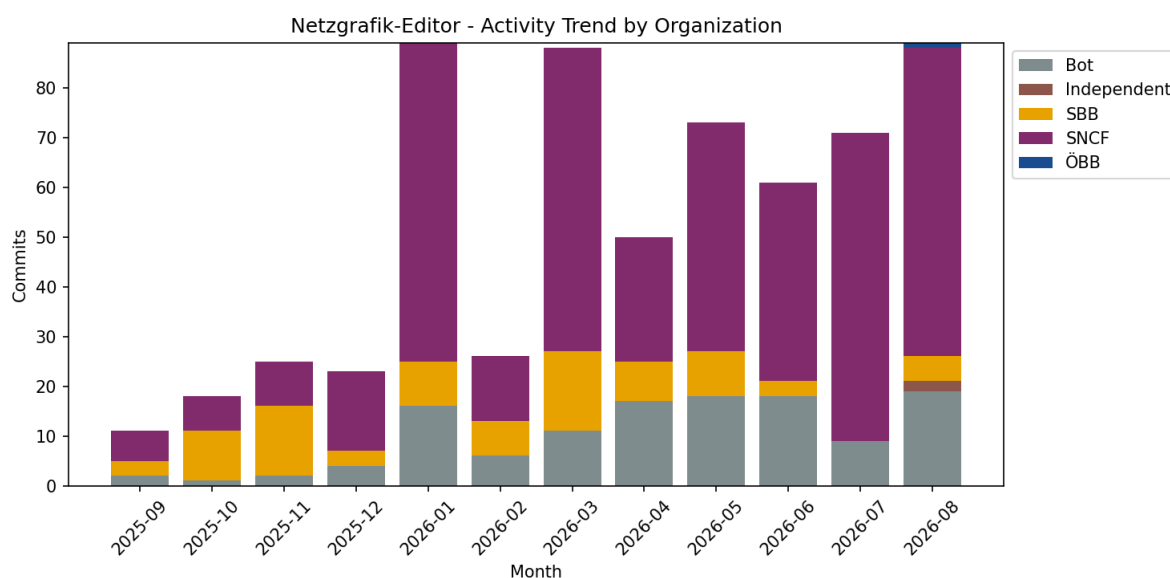
- [dac-migration-dss](#)
- [dac-migration-dss-popupsim](#)

Netzgrafik-Editor

Netzgrafik-Editor is a powerful software for creating, modifying, and analyzing regular-interval timetables. Originally developed by Swiss Federal Railways (SBB), it provides an intuitive graphical interface for timetable planning with real-time analysis capabilities, applicable across various transportation domains including rail, bus, and logistics planning.

12-Month Statistics:

- Total commits: **624**
- Total contributors: **23**
- Contributing organizations: **5**



Netzgrafik-Editor - 12-Month Activity Trend by Organization

Repositories:

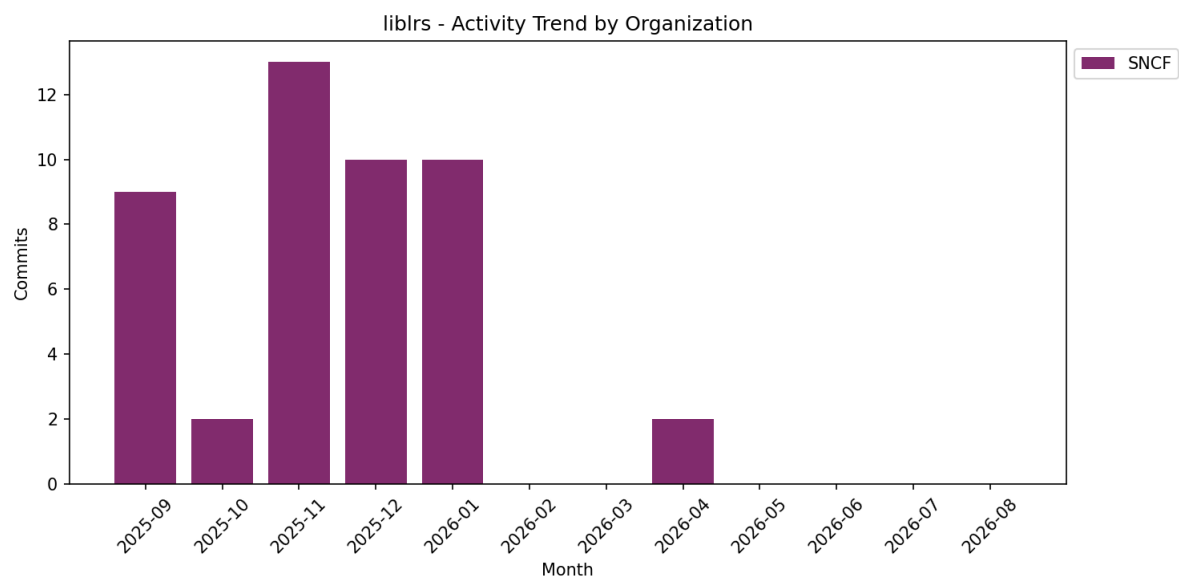
- [netzgrafik-editor-frontend](#)
- [netzgrafik-editor-backend](#)

liblrs

liblrs is a flexible, high-performance library for working with Linear Referencing Systems (LRS) and geographical data in the railway context. It enables manipulation of reference points along railway lines, handling complex edge cases like displaced anchors, varying distances, and multiple coordinate systems, with bindings for Rust, Python, and JavaScript.

12-Month Statistics:

- Total commits: **46**
- Total contributors: **4**
- Contributing organizations: **1**



liblrs - 12-Month Activity Trend by Organization

Repositories:

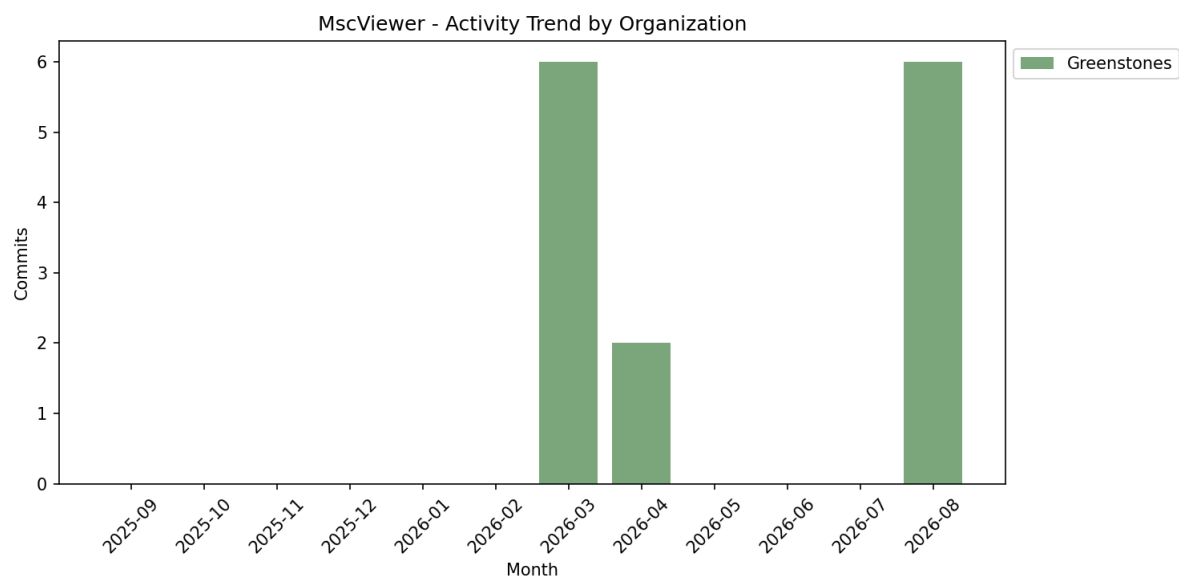
- [liblrs](#)

MscViewer

MscViewer is a toolkit for managing Nokia MSC/MSS (Mobile Switching Center) configurations through a modern web interface. It visualizes configuration hierarchies, relationships, and dependency graphs, renders configurations in geographical context on a map, and includes an AI agent for querying configurations in natural language and generating MSC/MSS commands. It is tailored to railway telecommunication needs such as GSM-R.

12-Month Statistics:

- Total commits: **14**
- Total contributors: **1**
- Contributing organizations: **1**



MscViewer - 12-Month Activity Trend by Organization

Repositories:

- [msc-viewer](#)

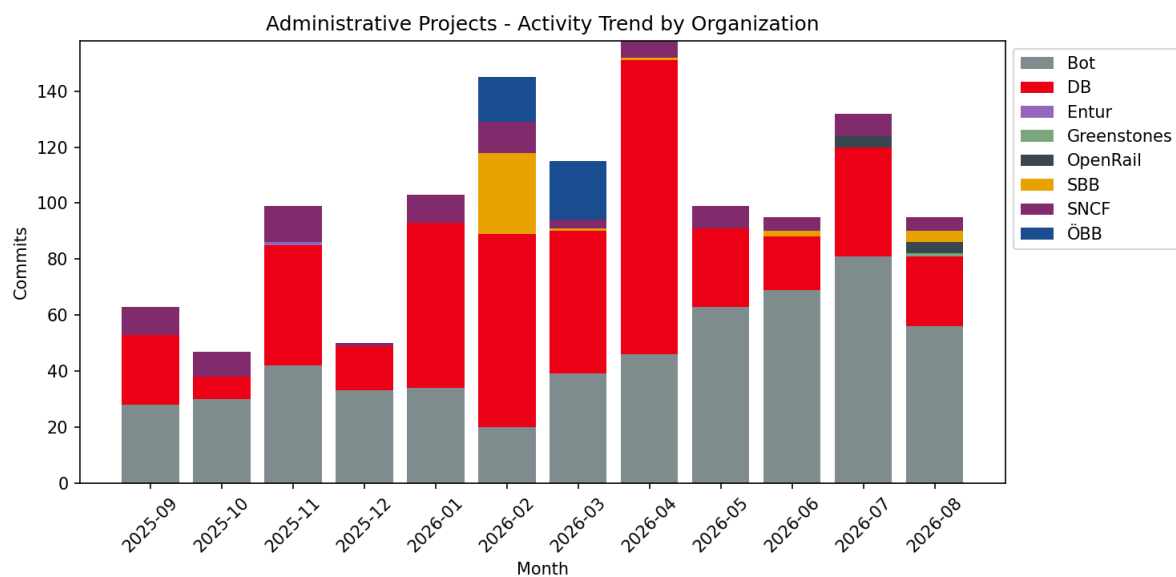
Administrative Projects

Administrative Projects

Administrative projects support the technical infrastructure and organizational operations of OpenRail Association. These include tools for GitHub management, compliance assistance, website maintenance, and conference organization, ensuring smooth operation of the association's technical ecosystem.

12-Month Statistics:

- Total commits: **1201**
- Total contributors: **31**
- Contributing organizations: **8**



Administrative Projects - 12-Month Activity Trend by Organization

Repositories:

- [annual-report](#)
- [authentik-user-manager](#)
- [compliance-assistant](#)
- [FOSDEM](#)
- [github-org-manager](#)
- [hack4rail-website](#)
- [landscape](#)
- [meetup](#)
- [nextcloud-forms-to-tables](#)
- [openrail-ai-toolkit](#)
- [openrail-ansible](#)
- [openrail-dns](#)
- [openrail-org-config](#)
- [project-template](#)
- [purl-tools](#)
- [renovate-config](#)
- [technical-committee](#)
- [web-deployment-action](#)
- [website](#)
- [wg-project-scouting](#)

Appendix

A. Methodology

This report is generated from Git commit history across all OpenRail project repositories. The metrics are calculated using the following definitions:

Contributors are individuals who have authored at least one commit in the Git history during the reporting period. Each contributor is identified by their email address and pseudonymized using SHA256 hashing to protect privacy.

Participants (not measured in this report) would include all individuals who actively engage with the projects, including contributors as well as those who create issues, comment on pull requests, or participate in discussions.

Organizations are determined through a manual mapping of contributor email addresses to their affiliated organizations. Contributors without a known organizational affiliation are categorized as “Unknown” and excluded from organization-specific metrics.

Automated commits from bots and CI/CD systems are filtered out based on commit metadata (merge commits and automated tooling commits are excluded).

Branch filtering: Only commits from main branches (main, master, or dev as specified per repository) are included in the metrics. Feature branches and unmerged work are excluded to focus on production-ready code.

All data is extracted from public Git repositories and aggregated without storing any personally identifiable information in the output reports.

B. About OpenRail

The OpenRail Association creates an open, autonomous and collaborative innovation space for the railway sector. Its mission is to promote the development of Open Source software projects in railways, similar to initiatives in telecommunications, automotive, and energy sectors.

The objective is to provide an open space to foster innovation and software development in the railway sector. Rather than isolated in-house applications, OpenRail federates IT teams around the world in open, collaborative, scalable and robust software projects. These are IT projects by the railways and for the railways.

OpenRail members include major railway organizations such as UIC, Deutsche Bahn, SNCF, SBB CFF FFS, Infrabel, ONCF, Entur, and the Flatland Association, all committed to the principles and benefits of Open Source for the railway industry.

C. Connect with Us

Website: <https://openrailassociation.org>

Email: contact@openrailassociation.org

GitHub: <https://github.com/OpenRailAssociation>

D. License



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